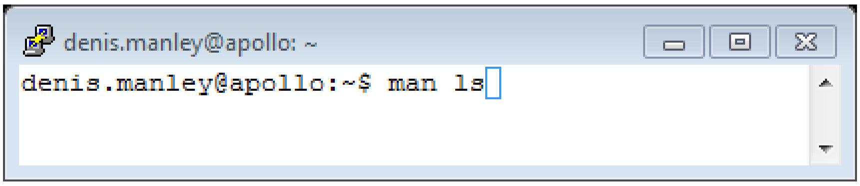
Operating Systems 2

Lab 1

This lab the student will create a directory in the linux operating system, change or move to the new directory and list its paths and contents. Write the "hello everyone I am " in c; Compile the code and run the code in the **command line linux environment**. Important *All programs in the module will be written, compiled and executed in the linux command line environment.*

******************* ceate directory ********************

The **ls** command is a very common function used in linux to list files, directories… :
Type the following at the prompt **man ls and press return**,



It display's what the command ls does and any other arguments that will modify what the default command does. .

press **q** (brings you back to the command prompt).

Command History –

If we press the **up-arrow key**, we will see that the previous command reappears after the prompt. This is called command history. Most Linux distributions remember the last five hundred commands by default. Press the **down-arrow** key and the previous command disappears.

TAB can be used to fill in the rest if a file name; e.g. vi H (press tab) results in vi HelloWorld.c (**this helps avoid misspelling of file names**)

Help is available - the man /info pages

- Get to know the system, use the **man pages** to get information on the following basic LINUX commands (the words in bold): commands. *These commands are used to create your first c program in a directory lab1*

- **Vi (basic linux text editor)**
- **mkdir** Make directory
- **pwd** the current path (directory)
- **ls** (list all files including directories)
- **cd** change directory

these is also a link to the VI “cheat sheet” which summaries the most useful commands associated with the **vi editor**:

Creating a directory:; navigate to the new directory

Clear the shell window : **clear**
Create a folder/directory in your account: **mkdir lab_1**
Change to this directory: **cd lab_1**
To see the absolute path: **pwd**

denis.manley@apollo: ~/lab1

```
denis.manley@apollo:~$ mkdir lab1
denis.manley@apollo:~$ pwd
/home/local/ICTDOMAIN/denis.manley
denis.manley@apollo:~$ cd lab1
denis.manley@apollo:~/lab1$ vi Helloworld.c
```

****************Create a c source file.; compile and run the file ***********

Create a c source file using vi editor **vi HelloWorld.c or nano HellowWorld.c**

Sample the C code HelloWorld program

denis.manley@soc-apollo: ~/OS2/week1

```
#include <stdio.h>

void main()

{

        /* this is a comment line
the printf display a string to the screen */

        printf("hello world from OS 2\n");

}    // (another way to insert comments) close braces
```

Compile the program HellowWorld.c

gcc –o helloworld HellowWorld.c

Run : **./helloworld** (note the *dot before the forward slash*)

```
denis.manley@soc-apollo:~/OS2/week1$ ./helloworld
hello world from OS 2
denis.manley@soc-apollo:~/OS2/week1$
```

Vi Editor instructions: Nano Instructions :refer to nano cheat sheet

Create a program vi HelloWorld.c (same rules as a variable name in C)

Press i to convert to insert mode: write text to vi editor

Type in the above code

Press esc to exit of out the insert mode to command mode

Enter **:wq** to save the file (:q to exit without saving) :

Question2
Write, compile and run programs discussed in lecture: **NestedWhile.c**